

THE HUMAN BODY 7



You use almost every muscle in your body when you roller-skate, but especially the leg and stomach muscles. Roller-skating normally burns about 500 calories an hour.

IN THIS UNIT, YOU WILL:

- Listen to a conversation about how humans are changing
- Watch a video about what happens when we lose one of our senses
- Watch or listen to a presentation about bacteria
- Survey classmates about the future
OR Discuss pros and cons of future situations

THINK AND DISCUSS:

1. Read the unit title. What do you want to learn about the human body?
2. Do you think the human body today is different from the human body 100 or 200 years ago? If yes, how?

EXPLORE THE THEME

Read the information. Then discuss the questions.

1. Do any of the facts surprise you?
2. How many parts of the body can you name?
3. Do you know or can you find any other interesting facts about the body?

The Amazing Human Body



1. You blink around 20 times a minute. That's about ten million times a year.



5. The word *muscle* comes from a Latin word that means "little mouse." The ancient Romans thought a bicep muscle looked like a mouse.



2. Scientists believe that the nose can recognize a trillion different scents!



6. Every year, you lose about 4 kg of dead skin cells.



3. Human teeth are as strong as shark teeth.



7. If you live to age 70, your heart will beat around 2.5 billion times!



4. Earwax is a type of sweat. Your ears make more earwax when you are afraid.



8. Bodies give off a small amount of light. It is too weak for the eye to see.



A Vocabulary

A MEANING FROM CONTEXT Read and listen to the article. Think about the meaning of the words in blue. 

National Geographic Explorer Christine Lee looks at a skeleton in the Museum of London.



THE MUSEUM OF LONDON

The Museum of London has **possibly** the largest collection of human **bones** in the world. The collection starts from over 5000 years ago and **continues** to the middle of the 19th century.

Most of the bones come from under the city. Every time there is new building work in the city, the museum knows the construction workers will almost **definitely** discover more objects from London's past—especially bones! These bones go to the museum. There, archaeologists use technology from modern **medicine**, such as X-ray machines, to study the bones.

Bioarchaeologists **certainly** learn a lot from the bones. For example, they can study the bones of someone from the 13th century and understand their **lifestyle**. Many people from that **period** worked in the fields. They **probably** spent **double** the amount of time on their feet that we do. As a result, their bones are much larger.

B Write each word in blue from exercise A next to its definition. TWO words have the same definition. Then listen, check, and repeat. 

1. _____ (n) the hard white parts inside a human or animal
2. _____ (adv) something is true; with no doubt
3. _____ (n) the way a person lives and works
4. _____ (adv) something might be true, but you are not sure
5. _____ (n) the study and science of health care
6. _____ (adv) something is likely to be true
7. _____ (n) a length of time with a beginning and an end
8. _____ (v) to not stop existing or happening
9. _____ (adj) twice as much or many

- **C** Cross out the **bold** words in each sentence and write a word from the box.

a bone continues double medicine lifestyle period

1. I fell off my bicycle and broke **my arm** _____.
2. My **routine** _____ is very stressful. I work all the time and never take time off.
3. My aunt is a doctor, and because of her, I'm thinking of a career in **this** _____.
4. Sorry I'm late. The trip took me **twice** _____ the time that I expected.
5. The first 20 years of your life is the **time** _____ when you grow the most.
6. The game **doesn't stop** _____ until one team has 17 points.

VOCABULARY SKILL Choose the Correct Meaning

When words have more than one meaning, a dictionary can help you choose the correct one for the context. A dictionary also gives you other information about the word.

form /fɔrm/ (n) **1** the shape of something: *The building is in the form of a T.*

2 a type, kind: *Yoga is a gentle form of exercise.* **3** a printed paper with spaces to be filled in: *When you apply for a visa, you must fill out a form.*

In this sentence, *form* has the first meaning because it refers to the shape of a human body.

*Bones and the skeleton create the **form** of the human body.*

- **D** Look at these definitions. Write the number of the correct definition next to each sentence below.

period /'pɪriəd/ n. **1** any length of time, long or short, that is part of a longer time: *We often have a rainy period in spring.* **2** a particular part of time in history: *The revolutionary period in American history was from 1775 to 1783.* **3** a regular division of time in a school day or a game: *Our lunch period is from 12:15 p.m. to 1 p.m.* **4** a punctuation mark or a dot ending a sentence: *This sentence ends with a period.*

- a. ____ I think you need a **period** here, not a question mark.
- b. ____ The year I spent in Japan was the happiest **period** in my life.
- c. ____ Jana has biology fourth **period** on Tuesdays and Thursdays.
- d. ____ The Renaissance was a **period** of growth in culture, art, and science.
- e. ____ My team were losing for a short **period** of the game, but they won in the end.
- f. ____ In my opinion, the best **period** for music was the 1960s.

- **E** Follow these steps.

1. Choose one of these words and look it up in a dictionary: *book, mean, medicine, object, or study.*
2. Choose one of its definitions. Write a sentence with that meaning. Circle the word.
3. Switch sentences with a partner. Look up the word your partner used. Find the correct definition.

A

Listening How Humans Are Changing

Critical Thinking



A COMPARE You are going to listen to a conversation about how humans are changing. Look at the photo. With a partner, make a list of differences between ancient humans and modern people. Think about clothing, food, transportation, work, activities, etc.

Ancient humans	Modern humans



B MAIN IDEAS Listen to the conversation between two students. Check (✓) the SIX types of change they discuss.

- | | | | |
|------------------------------------|--|------------------------------------|--|
| ☐ 1. height | ☐ 3. length of life | ☐ 5. bones | ☐ 7. brain size |
| ☐ 2. hair | ☐ 4. body temperature | ☐ 6. senses | ☐ 8. intelligence |

LISTENING SKILL Understand Time Periods

Listen for time expressions to help you understand the time period a speaker is talking about.

In the past

*a hundred years ago
during that period
in past centuries*

In the present

*nowadays
currently
at the moment*

In the future

*in one hundred years
in years to come
in the next few years/century*



C DETAILS Listen to the conversation again. For each topic, are the speakers referring to a change in the past, present, or future? TWO answers are correct for two of the topics.

	In the past	In the present	In the future
1. shorter humans	☐	☐	☐
2. better food and medicine	☐	☐	☐
3. longer life	☐	☐	☐
4. higher body temperature	☐	☐	☐
5. lower body temperature	☐	☐	☐
6. weaker humans	☐	☐	☐
7. larger brains	☐	☐	☐
8. double the population	☐	☐	☐



D FOCUSED LISTENING Listen to these sentences from the conversation. Write the exact numbers you hear.

- During that period, people only lived to be about _____ years old. Now, the average age is _____.

2. For example, women in many places will probably live to be _____ years old by the year _____.
3. He thinks humans will someday live to around _____ years!
4. In past centuries, the average human temperature was _____ degrees Fahrenheit. But at the moment, the average temperature of a human body is _____ degrees.
5. They think it will continue to fall by about, let me see, _____ degrees Fahrenheit every ten years.

CRITICAL THINKING Question and Check What You Hear

When you hear facts and figures, it's always important to question the information. Sometimes you can be sure the information is 100% true, but sometimes you are less certain. In fact, things you read or hear on the Internet may not be true at all. It's always a good idea to check the information in other sources, such as reliable websites or books in the library, or you could ask an expert.

- E** With a group, discuss the information from the conversation. For each statement below, decide if you all think it is:

Critical Thinking

- a. definitely true b. possibly true, but you're not sure c. probably not true

1. ____ The average human was a foot shorter a hundred years ago.
2. ____ People are taller nowadays because of better food and medicine.
3. ____ Someday, humans will live to around 130 years.
4. ____ Human body temperature is lower.
5. ____ People are weaker because we do less physical work.
6. ____ The human brain will continue to get smaller.
7. ____ Technology will make more decisions for us and it will become more intelligent.

- F** Where could you check the information in exercise E?
Discuss different sources with your group such as a website, a book, or an expert.

Ötzi, also called the Iceman, lived around 3300 BCE. This model of him is in the South Tyrol Museum of Archaeology in Bolzano, Italy.

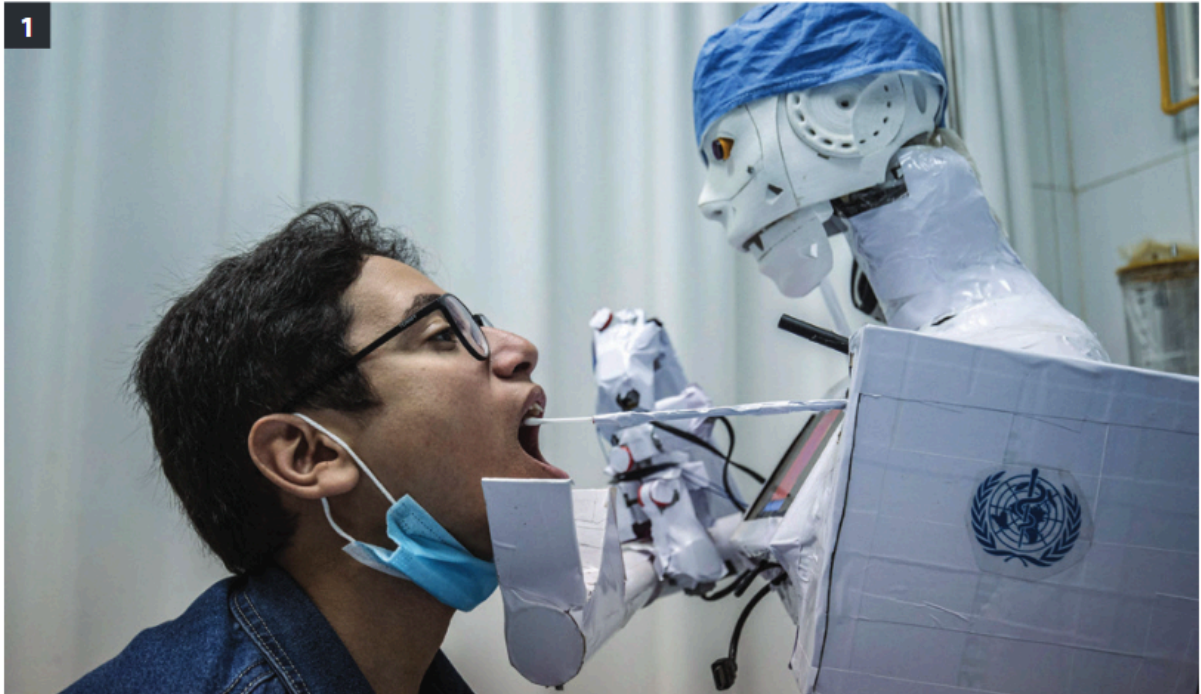


A

Speaking

Critical Thinking **A IDENTIFY** Look at the photos and discuss the questions as a class.

1. What technology does each photo show? How does the technology help people?
2. How do you feel about using technology in this way?



GRAMMAR FOR SPEAKING Will for Predictions; Adverbs of Certainty

Will for Predictions

We use *will* or *won't* + base verb to make predictions.

*In 20 years, robots **will be** doctors.*

*In the future, we **won't use** cell phones.*

*A: **Will** we **live** longer in the future?*

*B: Yes, we **will**. / No, we **won't**.*

*A: **Where will** you **live** in 20 years?*

B: I don't know!

Adverbs of Certainty: Possibly, Probably, Definitely, and Certainly

We can use adverbs of certainty after *will* and before *won't*.

*Robots **will probably (not) be** doctors someday.*

*I **definitely won't become** a doctor.*

We can also use adverbs of certainty to reply in conversations.

*A: **Will** you **become** a schoolteacher?*

B: Possibly. Or I'll work in a university.

B Work with a partner and take turns. Unscramble the predictions and say the sentences.

1. find / a cure / will / scientists / for cancer
2. live / humans / won't / other planets / on
3. computer technology / will / have / inside / our bodies / we
4. most of / machines and robots / will / do / our work
5. humans / won't / physical exercise / a lot of / do
6. will / there / one / be / world language
7. people / cars / won't / to travel / anymore / use
8. will / teach / all universities / online / not in buildings

C Choose FOUR predictions from exercise B. How much do you agree or disagree with them? Rewrite them *with will/won't* and an adverb of certainty.

1. _____
2. _____
3. _____
4. _____

D ARGUE Work with a partner. Share your sentences from exercise C and compare your predictions. Give reasons for your predictions.

A: Scientists will probably find a cure for cancer.

B: I disagree. In my opinion, they probably won't find a cure for cancer. . . .

Critical Thinking

SPEAKING SKILL Talk about Possibilities

We can use these adverbs and expressions to talk about possibilities. The language we choose shows how certain we are that something is true.

Ask about possibilities

How certain are you ... ?

Is it possible that you will ... ?

Do you think that ... ?

How likely is it that ... ?

Respond

Definitely. / I'm (very) certain that ...

Probably. / I'm fairly sure that ...

Maybe. / It's possible, (but ...)

I doubt that ...

100%

75%

50%

25%

E Match the questions to the responses. Then practice the conversations with a partner.

- | | |
|--|---|
| 1. How certain are you about your future career? _____ | a. Definitely. I love visiting new places and learning new languages. |
| 2. Is it possible that you will become a doctor? _____ | b. Maybe. I'm interested in medicine, and I like helping people. |
| 3. Do you think you'll live in other countries someday? _____ | c. I'm fairly sure that I want to be a nurse. |
| 4. How likely is it that you will do the same job for your whole life? _____ | d. Oh, I doubt that I'll do that. I like change! |

F PERSONALIZE Work with the same partner and take turns. Ask the four questions in exercise E. Give your own answers.

Critical Thinking

G RATE Read the questions. How possible is each situation? Write a percentage in the chart. For example: 50% (= *It's possible, but ...*). Then write one more question of your own.

Future possibilities	You	Your partner
1. In the future, will we have small cell phones inside our brains?	%	%
2. Is it possible that students will stay in school until they are 25?	%	%
3. How likely is it that humans will stop eating meat?	%	%
4. Do you think artificial intelligence will replace artists?	%	%
5. Is it possible that there will be no countries or borders someday?	%	%
6.	%	%

H Work with a partner and take turns. Ask the questions from exercise G. Respond and discuss possibilities. Give reasons for your answers.

A: In the future, will we have small cell phones inside our brains?

B: No, I doubt we will ever have phones in our brains.

How certain is your partner? Write a percentage for them. For example: *Yes, I'm fairly sure. = 75%*

Brain Connections

accident (n) an unexpected bad event, such as a car crash
connection (n) when two things are joined together

scan (v) to look inside something with a machine
active (adj) working in a normal way

▲ This graphic shows how the body sends information about the world to the brain.

● **A** Watch the video and complete these sentences with ONE word. ▶

1. Senses send information to different parts of the _____.
2. When a sense is not very strong, we can use _____ to help us.
3. When we lose a sense, connections in the brain get _____.
4. In addition, _____ connections can grow in the brain.

● **B** Which sentences are true and which are false? Choose T or F. Then watch the video and check your ideas. Correct the false sentences. ▶

- | | | |
|---|---|---|
| 1. There are six basic senses. | T | F |
| 2. The brain receives information from the senses. | T | F |
| 3. People can lose a sense over long periods of time or suddenly. | T | F |
| 4. A blind person's hearing can improve. | T | F |
| 5. In deaf people, the part of the brain for hearing is not active. | T | F |
| 6. Scientists now know almost everything about the brain. | T | F |

C PERSONALIZE Which do you think is your strongest and most important sense? Why? Discuss your ideas with a partner.

B

Vocabulary

A Listen and repeat. Check (✓) any words you already know. 

benefit (n)

cause (v)

disease (n)

improve (v)

mental (adj)

blood (n)

control (v)

feed (v)

lead to (v phr)

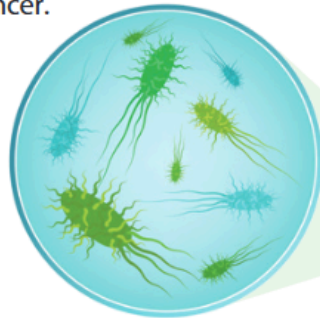
protect (v)

B **MEANING FROM CONTEXT** Listen and write the correct form of the words from exercise A. Then think about the meaning of each word. 

THE ¹ _____ OF GOOD BACTERIA

Bad bacteria in our bodies can

² _____ illness. However, most bacteria in our stomach is good, and it ³ _____ the human body from ⁴ _____ such as cancer.



Bacteria in your stomach helps

⁵ _____ the amount of sugar in your ⁶ _____.

A healthy stomach also

⁷ _____ better

⁸ _____

health—you feel happier.

You can ⁹ _____

the bacteria in your body.

¹⁰ _____ it

healthy food like fruit, olive oil, and dark chocolate, and don't eat too much processed food.



-
- **C** First, choose the correct words to complete this quiz. Then work with a partner and take the quiz. How many questions can you answer correctly?

QUIZ: HEALTH AND THE HUMAN BODY

1. How many liters of (**blood** / **disease**) are in your body?
 - a. about 3
 - b. about 5
 - c. about 7
2. Which part of the body (**controls** / **feeds**) the movement of your blood?
 - a. stomach
 - b. brain
 - c. heart
3. Which part of the body controls your (**mental** / **memorable**) health?
 - a. stomach
 - b. brain
 - c. blood
4. Which activity (**leads to** / **improves**) your mental health?
 - a. any type of physical exercise
 - b. looking at social media
 - c. eating fast food
5. How many different types of stomach bacteria (**protect** / **cause**) your body?
 - a. more than one hundred
 - b. more than a thousand
 - c. more than a million
6. To promote good bacteria, which should you do every day?
 - a. (**Feed** / **Control**) the bacteria a lot of fruit.
 - b. Buy food with bacteria in it.
 - c. Don't exercise too much.
7. What is cancer?
 - a. a type of blood
 - b. a type of bacteria
 - c. a type of disease

- D** **CREATE** Work with a partner. Write two more quiz questions about the human body (with a, b, c answers). Then join another pair and take turns. Ask and answer your questions.

Critical Thinking

B

Listening The Benefits of Bacteria

Critical Thinking

A PREDICT You are going to listen to a presentation about the benefits of bacteria. Work with a partner. Look at the three sets of words. Try to predict what the lecturer will say about these words. What is the connection between the words?

bacteria
bad protect
disease good

bacteria
stomach
sugar
blood

bacteria
healthy eating
sleep
physical exercise

> *Bad bacteria can lead to diseases...*

B MAIN IDEAS Listen to the first part of the presentation. Check (✓) the THREE topics the lecturer plans to talk about. 

1. ☐ everything your body needs for good health
2. ☐ benefits of good bacteria
3. ☐ similarities between good and bad bacteria
4. ☐ bacteria in your stomach
5. ☐ how to help the good bacteria in your body
6. ☐ how to make bad bacteria into good bacteria

C Watch or listen to the presentation. Take notes about the words in exercise A. Were your predictions about the words correct?  

NOTE-TAKING SKILL Use an Outline

When you listen to a lecture or presentation, you can organize your notes using an outline. A basic outline includes the main ideas and supporting details. You can use letters and numbers to organize the main ideas and supporting details.

Bacteria in the stomach

- **D DETAILS** Watch or listen again. Complete the notes with ONE or TWO words.  

The Benefits of Bacteria

1. Difference between good and bad bacteria
 - a. Bad bacteria causes different _____ in humans and animals.
 - b. Good bacteria is important for the _____ and health
2. Bacteria in stomach
 - a. Good stomach bacteria _____ from diseases.
 - b. Bacteria in adult stomachs can prevent _____ like _____.
 - c. Stomach bacteria helps control _____ in your blood.
 - d. Some scientists believe stomach bacteria sends positive _____ to your _____.
3. Help bacteria with a healthy lifestyle.
 - a. Don't eat the _____ every day.
 - b. Food for good bacteria = fruit, yogurt, olive oil, _____
 - c. Sleep 7 to 8 hours a night at _____.
 - d. Do 30 minutes of _____ every day.

PRONUNCIATION Recognize Reduced Forms

 In everyday speech, some speakers reduce certain verb forms, such as *will* and *going to*. It's important to recognize these forms so you can understand a speaker's ideas.

You read or write:

*Bad bacteria **will** cause diseases.*

*But we're **going to** look at the benefits.*

You may hear:

*Bad **bacteria'll** cause disease.*

*But we're **gonna** look at the benefits.*

- **E** Listen to sentences from the presentation that have reduced forms. Write the full forms of the TWO or THREE missing words. Then practice the sentences and try to say the reduced forms. 

1. Today _____ talk about something your body really needs.
2. _____ explore the benefits of bacteria.
3. So, a healthy _____ send positive messages to your brain.
4. And _____ make you happier.
5. Now _____ tell you how to improve your stomach bacteria.

- F PERSONALIZE** Think about the presentation and discuss these questions with a partner.

1. Before this presentation, did you think all bacteria were bad?
2. Do you think your brain is happier when your stomach is happy?

GRAMMAR FOR SPEAKING *Will and Be Going To*

We can use *will* or *be going to* + base verb to **make predictions** about the future.

*I think you'll **get** a job as a doctor in the future.*

*I think you're **going to be** a doctor when you grow up.*

We usually use *be going to* + base verb to **talk about future plans or intentions**.

*I'm **going to study** medicine. Then I'm **going to work** in a hospital.*

We usually use *will* + base verb to **make a decision at the time of speaking** (such as a promise to do something).

A: I don't understand this homework.

*B: Don't worry. I'll **help** you.*

- A** Choose the correct verb forms in these conversations. Both verb forms are possible in ONE conversation. Listen and check. Then practice the conversations with a partner. 

- A: I have to complete this college application. It's so hard!
B: Don't worry. I ('ll help / 'm going to help) you.
A: Really? That'd be great.
- A: What (will you study / are you going to study) in college?
B: Biology. I ('ll be / 'm going to be) a doctor.
A: That's fantastic! Good luck!
- A: What time do you finish work at the hospital tonight?
B: At midnight! I ('ll be / 'm going to be) so tired.
A: Definitely!
- A: Hi Marek. I'm at the train station. How do I get to your house?
B: Hi Sarah. Stay there. I ('ll / 'm going to) drive there and meet you.
A: Oh, thanks!

Critical Thinking

- B ANALYZE** Think about the uses of *will* and *be going to* in exercise A. In which conversation(s) (1-4) does a person:

- _____ make a prediction?
- _____ talk about a future plan or intention?
- _____ make a decision at the time of speaking?

Critical Thinking

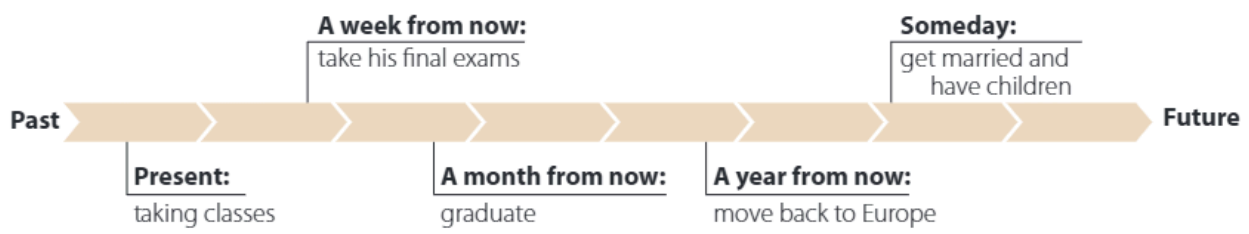
- C CREATE** Work with a partner. Write three short conversations with these sentences. Then join another pair and present your conversations.

- One day I'm going to do it!
- Don't worry. I'll pay for it.
- It'll probably arrive tomorrow morning.



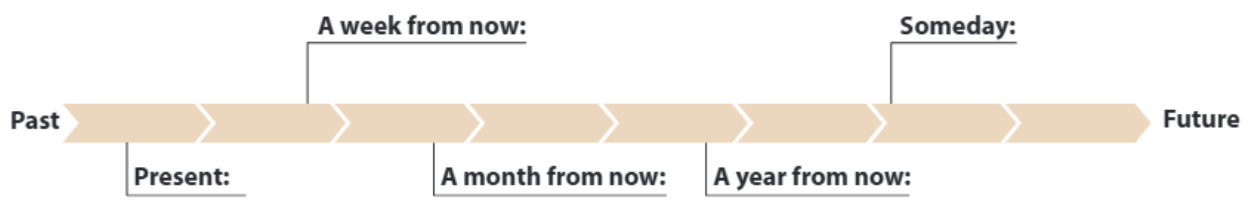
Students celebrate their graduation in Costa Mesa, California, USA.

D Look at the timeline that shows John's plans. Complete the sentences below with the correct verb forms. Then share your sentences with a partner.



1. Right now, John _____ at college.
2. In a week, he _____.
3. Then in a month, he _____.
4. In a year, John _____ where his family lives.
5. Someday in the future, perhaps John _____.

E PERSONALIZE Complete this timeline for you. Write your plans.



F Work with a group. Take turns. Present and describe your timeline.

> *Right now, I'm at school, but in a week, I'm going to go on vacation!*

Review

SELF-ASSESS

How well can you . . . ?	Very well.	OK.	I need improvement.
use the key vocabulary	☐	☐	☐
recognize reduced forms	☐	☐	☐
use <i>will</i> and <i>be going to</i>	☐	☐	☐
talk about possibilities	☐	☐	☐

A VOCABULARY Choose the correct word in each sentence.

- I always (feed / cause) my cat at eight o'clock.
- There's (bone / blood) on your face. Did you cut yourself?
- You have a very healthy (lifestyle / disease).
- Regular physical exercise can (improve / lead to) your health.
- What's the main (period / benefit) of good bacteria?

B PRONUNCIATION Listen to sentences using reduced and full forms. Check (✓) the sentence you hear.

- ☐ English lessons'll start tomorrow.
 - ☐ English lessons will start tomorrow.
- ☐ We're gonna pass this exam.
 - ☐ We're going to pass this exam.
- ☐ What'll you do after school?
 - ☐ What will you do after school?
- ☐ My daughter's gonna study biology.
 - ☐ My daughter's going to study biology.

C GRAMMAR Find and correct the mistake in each sentence.

- I'll to visit the doctor tomorrow.
- We won't probably go out tonight.
- She going to be a doctor.
- I promise I'll calling you back later.
- What you going to do when you leave college?

D SPEAKING SKILL Say answers to these questions. Use expressions from the box and include reasons.

I'm very certain that . . . Probably. I'm fairly sure that . . . I doubt that . . . Maybe.

- How certain are you that you'll pass all your exams this year?
- Is it likely that you'll have more than one career in the future?

RE-ASSESS What skills or language do you still need help with?

OPTION 1 Survey classmates about the future

- A** Interview your classmates about their futures. Use the words in the survey. If they answer, "Definitely / probably / possibly . . .," ask for a reason and write your classmate's name.

A: Do you think you'll start a new kind of exercise this year?

B: Probably, because I want to get in better shape.

If they answer, "No, I (definitely / probably / possibly) won't," ask another person. You have ten minutes. Try to write names in every square.

start a new kind of exercise this year	go to college	learn to drive a car	move to a new home in the next five years
pass your exams this year	go camping for a vacation	learn another language someday	do some volunteer work this year
write a book someday	become famous someday	learn to make a new dish this week	travel to another country in the next year
go out with friends this weekend	try a new musical instrument	become a vegetarian someday	run a marathon

▼ Yoga is good physical exercise, and it helps decrease stress.



OPTION 2 Discuss pros and cons of future situations

- A MODEL** Listen to a group discussion and answer the questions. Choose one or more answers for each question. 

1. What topic(s) does Yoko introduce?
 - a. Robots will do our jobs.
 - b. Robots will drive cars.
 - c. Robots will work in factories.
2. What pros does Hasan talk about?
 - a. Robots are better at everyday jobs.
 - b. Humans won't have to do boring jobs.
 - c. Humans will have more free time.
3. What cons does Claudia describe?
 - a. Humans will get bored.
 - b. Humans won't earn money.
 - c. Robots will stop working sometimes.

COLLABORATION SKILL Participate in a Group Discussion

Here are some things you should remember to do in an academic or work discussion.

As an individual, make sure you:

- give your honest opinion
- listen carefully
- agree or disagree politely
- encourage others

As a group, make sure everyone:

- understands the goal
- participates

 **ONLINE** If you're having trouble keeping a conversation going, you can agree to pass the conversation. After you finish speaking, name someone else to take a turn.

- B PLAN** Complete the chart. How likely do you think each statement is? Write *definitely*, *probably*, *maybe*, or *probably not*. Then write one pro and one con for each prediction.

Statement	How likely?	Pros	Cons
Drones will transport everything to people's homes, such as shopping and medicine.			
You'll always meet human doctors online, and there will be robots in hospitals.			
The average human will live to 150 years of age or longer.			

- C DISCUSS** Work with a group of three. Each group member introduces one of the statements from exercise B and asks the others for ideas. Answer these questions:

- How likely is each future situation?
- What are the pros and cons of each situation for people and communities?
Are there more pros or more cons?

- D REPORT** As a group, report back to the class about one of the topics you discussed. How likely is the prediction? How do you think it could affect the world?